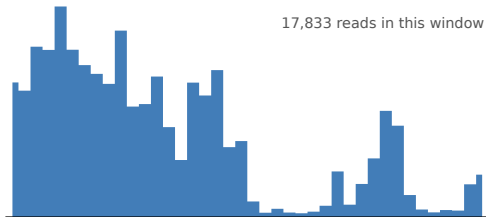
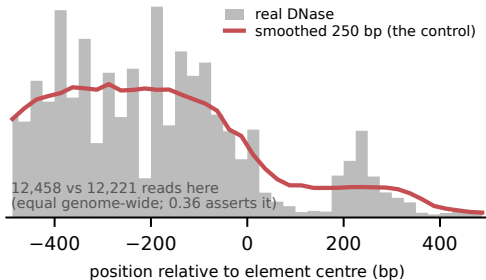


a

Real ATAC



Real DNase, and the control built from it

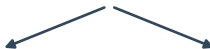
**b**

Why does real DNase beat real ATAC as the model's input?

Two properties differ at once, and they have opposite implications for the converter.



Feed the model DNase smoothed at 250 bp:
magnitude kept, shape removed



advantage
SURVIVES

advantage
COLLAPSES

shape never mattered.
The converter's strength is
worth nothing here; only
better counts could help.

shape is what DNase was
winning on, which is exactly
what the converter is
good at.

250 bp is chosen, not arbitrary: it is where observed ATAC and DNase already agree about shape ($r = 0.92$ at 250 bp against 0.29 at 1 bp), so smoothing there removes close to exactly the information ATAC does not already carry.